

Lecture 3

Overview of Archaea and Bacteria



Dr Philip Button



Approaches to Classifying Organisms

A two-kingdom system—Linnaeus



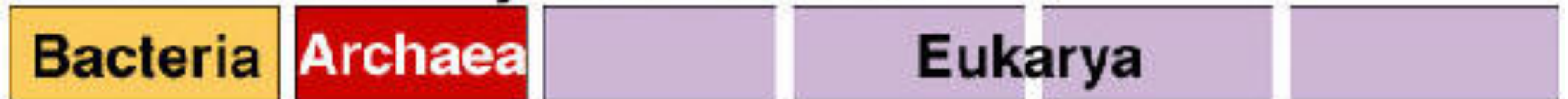
A five-kingdom system—Whittaker



A six-kingdom system—Woese



A three-domain system—Woese



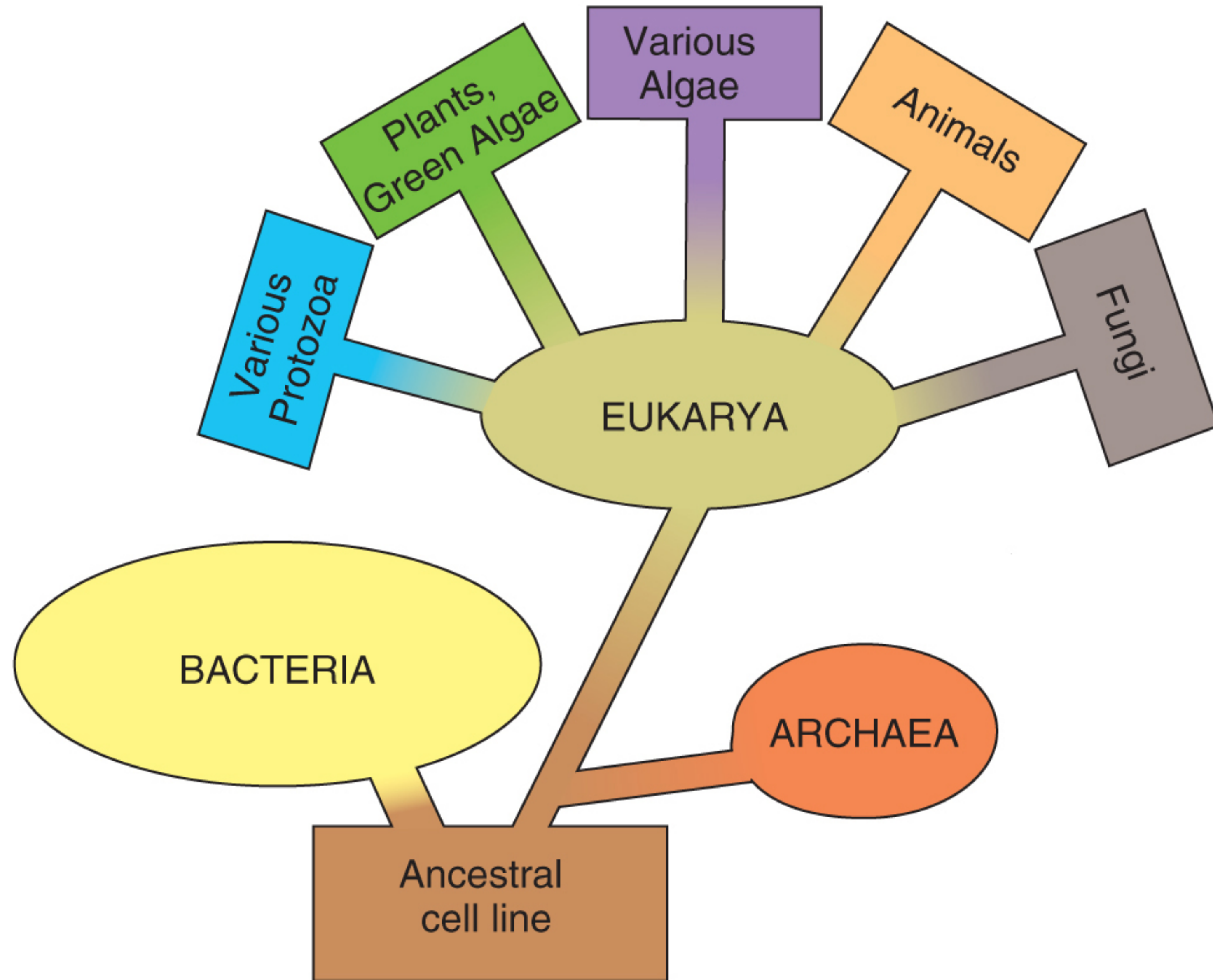


TABLE 4.3 Comparison of Three Cellular Domains

Characteristic	Bacteria	Archaea	Eukarya
Cell type	Prokaryotic	Prokaryotic	Eukaryotic
Chromosomes	Single, or few, circular	Single, circular	Several, linear
Types of ribosomes	70S	70S but structure is similar to 80S	80S
Contains unique ribosomal RNA signature sequences	+	+	+
Number of sequences shared with Eukarya	1	3	(all)
Protein synthesis similar to Eukarya	—	+	
Presence of peptidoglycan in cell wall	+	—	—
Cell membrane lipids	Fatty acids with ester linkages	Long-chain, branched hydrocarbons with ether linkages	Fatty acids with ester linkages
Sterols in membrane	— (some exceptions)	—	+



Example of Archaea

Genus	Optimal temperature	Optimal pH
<i>Thermoplasma</i>	57	1.5
<i>Picrophilus</i>	60	0.7
<i>Sulfolobus</i>	75	2.5
<i>Desulfurococcus</i>	85	6.0
<i>Methanopyrus</i>	98	7.0
<i>Pyrodictium</i>	105	6.0

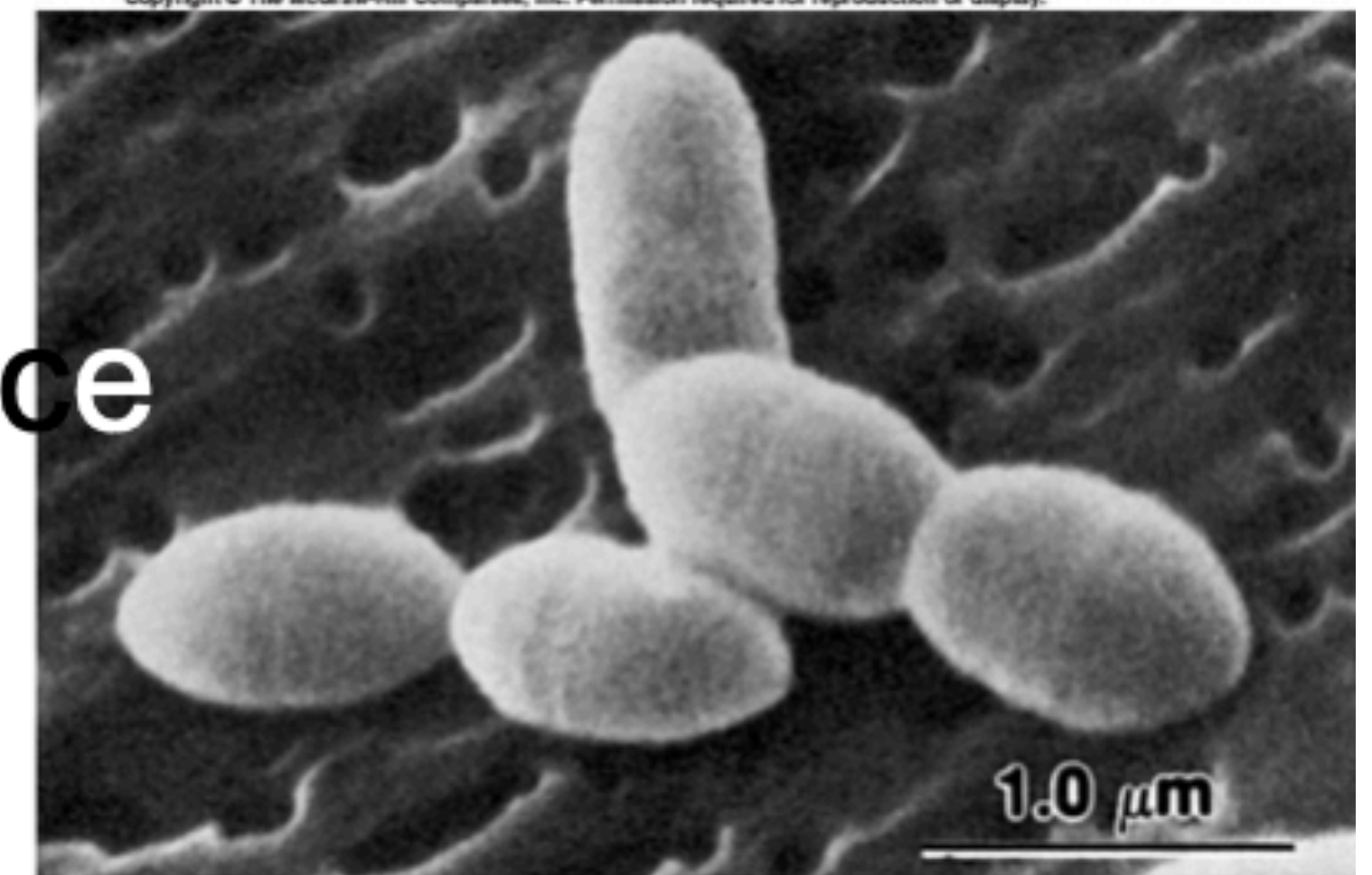
Halobacteriales

- 17 genera in one family
- Minimum 1.5 M NaCl for growth
- Optimal NaCl concentration for growth is 3-4 M



Methanogenic Archaea

- Strict aerobes
- Form CH_4 or CH_4 and CO_2 from a variety of compounds
- Largest group of Archaea
 - Five orders and 26 genera
- Potential industrial relevance
 - Clean burning fuel
 - Excellent energy source



(a)

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Domain

Bacteria

Phylum

Proteobacteria

Class

α -proteobacteria

β -proteobacteria

γ -proteobacteria

δ -proteobacteria

ϵ -proteobacteria

Order

Chromatiales

Thiotrichales

Legionellales

Pseudomonadales

Vibrionales

Enterobacteriales

Pasteurellales

Family

Enterobacteriaceae

Genus

Enterobacter

Escherichia

Klebsiella

Proteus

Salmonella

Serratia

Shigella

Yersinia

Species

S. boydii

S. dysenteriae

S. flexneri

S. sonnei





Figure 40.12

20 *Bifidobacterium* sp.

